Pairs of forms in geometric mechanics

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Introduction

- There are many relevant geometric structures in mechanics formed by a **1-form** and a **2-form**. For example, (co)symplectic, contact, locally conformally symplectic, etc...
- The dynamics arise from a vector field associated to the differential forms (Hamiltonian vector fields, Reeb vector fields...).
- Our aim is to provide a general framework describing the fundamental properties of these pairs of forms.

Rank of a 1-form

Given a p-form $\alpha \in \Omega^p(M)$, its **kernel** at a point $x \in M$ is

$$\operatorname{Ker} \alpha_x = \{v \in \mathrm{T}_x M \mid \iota_v \alpha_x = 0\} .$$

The **rank** of α is

$$\operatorname{rank} \alpha_x = \operatorname{codim} \operatorname{Ker} \alpha_x = \dim \left(\operatorname{Ker} \alpha_x\right)^\circ .$$

- If $p = 1$, then $(\operatorname{Ker} \alpha_x)^\circ = \langle \alpha_x \rangle$, so the rank is 1 if $\alpha_x \neq 0$ and zero otherwise.
- If $p = 2$, the rank is even at every point. Also, this rank is $2r$ at a point x if and only if

$$(\alpha)^r \neq 0 , \qquad (\alpha_x)^{r+1} = 0 .$$

Class of a pair

Definition

The **characteristic distribution** of a pair $\tau \in \Omega^p(M), \omega \in \Omega^q(M)$ at $x \in M$ is

$$\mathcal{K} = \mathrm{Ker}\, \tau_x \cap \mathrm{Ker}\, \omega_x \,.$$

The **class** of a pair (τ, ω) at x is

$$\mathrm{cl}(\tau, \omega)_x = \mathrm{codim}\, \mathcal{K}_x = \dim \mathcal{K}_x^\circ \,.$$

Note that

$$\mathcal{K}_x^\circ = (\mathrm{Ker}\, \tau_x \cap \mathrm{Ker}\, \omega_x)^\circ = (\mathrm{Ker}\, \tau_x)^\circ + (\mathrm{Ker}\, \omega_x)^\circ \,.$$

Doublets

From now on, we will consider a pair (τ, ω) such that $\tau \in \Omega^1(M)$ and $\omega \in \Omega^2(M)$.

Definition

A pair (τ, ω), with $\tau \in \Omega^1(M)$ and $\omega \in \Omega^2(M)$, is a **doublet** if the class is constant.

- If $\mathrm{cl}(\tau, \omega)$ is $2r$ or $2r+1$, then the rank of ω is necessarily $2r$.
- If the class is odd then, necessarily, the rank of τ is one (so $\tau_x \neq 0$).
- If the class is even, the rank of τ may not be constant.

Reeb and Liouville vector fields

The following vector fields are related to the parity of the class of a doublet.

Definition

Let (τ, ω) be a pair of a 1-form and a 2-form.

- A **Reeb vector field** for the pair is a vector field R satisfying

$$\iota_R \tau = 1 \,, \qquad \iota_R \omega = 0 \,.$$

- A **Liouville vector field** for the pair is a vector field Δ satisfying

$$\iota_\Delta \omega = \tau \,.$$

Class and distinguished vector fields

The parity of the class gives necessary and sufficient conditions for the existence of Reeb or Liouville vector fields.

Theorem

Let (τ, ω) be a doublet.

- *The following statements are equivalent:*
  1. *The class of the doublet is* **odd**.
  2. $\operatorname{Ker} \omega \not\subset \operatorname{Ker} \tau$.
  3. *The doublet has a* **Reeb** *vector field.*

- *The following statements are equivalent:*
  1. *The class of the doublet is* **even**.
  2. $\operatorname{Ker} \omega \subset \operatorname{Ker} \tau$.
  3. *The doublet has a* **Liouville** *vector field.*

Algebraic conditions

Proposition

The pair (τ, ω) is a **doublet of class 2r+1** *if, and only if,*

$$\omega^{r+1} = 0\,, \qquad \tau \wedge \omega^r \neq 0\,.$$

Proposition

The pair (τ, ω) is a **doublet of class 2r** *if, and only if,*

$$\omega^r \neq 0\,, \qquad \omega^{r+1} = 0\,, \qquad \tau \wedge \omega^r = 0\,.$$

If, additionally, τ is nowhere-vanishing, then $\mathrm{cl}(\tau, \omega) = 2r$ iff

$$\omega^r \neq 0\,, \qquad \tau \wedge \omega^r = 0\,.$$

Characteristic tensor field

Definition

The **characteristic tensor field** associated with a pair (τ, ω) is the 2-covariant tensor field given by

$$B = \tau \otimes \tau + \omega\,.$$

In general B is not symmetric nor skew-symmetric. This tensor field defines a vector bundle morphism,

$$\begin{array}{rccc} \widehat{B}\colon & \mathrm{T}M & \longrightarrow & \mathrm{T}^*M \\ & v & \longmapsto & B(v,\cdot) = \iota_v B = (\iota_v\tau)\tau + \iota_v\omega\,. \end{array}$$

It can be extended to the modules of sections of these bundles.

Characteristic tensor and class

Proposition

Let (τ, ω) be a pair of a differential 1-form and a 2-form. Then,

$$\mathcal{K} = \operatorname{Ker} \tau \cap \operatorname{Ker} \omega = \operatorname{Ker} \widehat{B} = (\operatorname{Im} \widehat{B})^{\circ} \,.$$

So the **class of a pair equals the rank of** $\widehat{B}$.

Hence, the following conditions are equivalent:

1. $\widehat{B}$ is a vector bundle isomorphism.
2. The characteristic distribution of the pair is zero.
3. The class of the pair equals the dimension of the manifold.

Reeb and Liouville v.f. are characterized as the elements of $\widehat{B}^{-1}(\tau)$, i.e.

- **Even class:** $B(X) = \tau \iff$ X is a **Liouville** vector field,
- **Odd class:** $B(X) = \tau \iff$ X is a **Reeb** vector field .

Presymplectic forms

Definition

A **presymplectic structure** on a manifold M is a 2-form ω such that

- ω is closed, and
- the rank of ω is constant.

In this case, we can consider the doublet $(0, \omega)$, so $\mathrm{cl}(0, \omega) = \operatorname{rank} \omega = 2r$. We have $B = \omega$ and the Liouville v.f. of the pair are the sections of $\operatorname{Ker} \omega$.

If ω is exact, with $\omega = \mathrm{d}\theta$, then the pair (θ, ω) does not necessarily have constant class. Even when the class is constant, it might be even or odd.

If ω is symplectic and exact, (θ, ω) has class $2n$, and the map

$$\widehat{B}(X) = \iota_X \omega + (\iota_X \theta)\theta \,,$$

is an isomorphism. The Liouville vector field is $\Delta = \widehat{B}^{-1}(\theta) = \widehat{\omega}^{-1}(\theta)$.

Hamiltonian systems

Definition

A **Hamiltonian system** is a triple (M, ω, H), where M is a manifold, $\omega \in \Omega^2(M)$ is a presymplectic form and H is a function.

The **Hamiltonian v.f. are the Liouville v.f.** of the pair $(\mathrm{d}H, \omega)$

$$\iota_X \omega = \mathrm{d}H\,.$$

If ω is symplectic, the Hamiltonian vector field exists and is unique. If ω is degenerate, we need to solve

$$\iota_{X_x} \omega_x = \mathrm{d}_x H\,,$$

which has a solution iff $\mathrm{d}_x H \in \operatorname{Im} \widehat{\omega}_x$. Equivalently, $(\mathrm{d}H, \omega)$ has even class at x. In the language of constrained systems: *the primary constraint submanifold is the set of points where the class of* $(\mathrm{d}H, \omega)$ *is even.*

Contact forms

Definition

A **contact form** on a manifold M of dimension $2n+1$ is a 1-form η such that

- $\eta \wedge (\mathrm{d}\eta)^n$ is a volume form on M, i.e. it is nowhere-vanishing.

This is equivalent to the pair $(\eta, \mathrm{d}\eta)$ being a doublet of class $2n+1$. So the characteristic distribution $\mathcal{K} = \operatorname{Ker}\eta \cap \operatorname{Ker}\mathrm{d}\eta$ is zero.

The characteristic tensor field defines an isomorphism

$$\widehat{B}(X) = \iota_X \mathrm{d}\eta + (\iota_X \eta)\eta \,.$$

Since the class is odd, there exists a Reeb vector field R, which is uniquely defined as $R = \widehat{B}^{-1}(\eta)$.

Precontact forms

Definition

A 1-form η on a manifold is a **precontact form** if

- η is nowhere-vanishing, and
- the pair $(\eta, \mathrm{d}\eta)$ has constant class.

1. η is a precontact form of class $2r+1$ if, and only if,

$$\eta \wedge (\mathrm{d}\eta)^r \neq 0\,, \qquad (\mathrm{d}\eta)^{r+1} = 0\,.$$

2. η is a precontact form of class $2r+2$ if, and only if,

$$\eta \neq 0\,, \qquad \eta \wedge (\mathrm{d}\eta)^{r+1} = 0\,, \qquad (\mathrm{d}\eta)^{r+1} \neq 0\,.$$

Example of precontact form

The following example shows a precontact form of even class coming from the study of singular Lagrangian functions within contact mechanics.

Example

Consider the manifold $M = \mathrm{T}^*(\mathbb{R}^3) \times \mathbb{R}$ and the 1-form

$$\eta = \mathrm{d}s - p_x \mathrm{d}x + \gamma s \mathrm{d}y - p_z \mathrm{d}z \,,$$

where $\gamma \in \mathbb{R}$ is a non-zero constant. One can check that

$$\operatorname{Ker} \mathrm{d}\eta = \langle \partial/\partial p_y \rangle \subset \operatorname{Ker} \eta \,,$$

so $\mathcal{K} = \operatorname{Ker} \eta \cap \operatorname{Ker} \mathrm{d}\eta = \left\langle \frac{\partial}{\partial p_y} \right\rangle$, and the class is 6. The Liouvilles are

$$\Delta = -\frac{1}{\gamma}\frac{\partial}{\partial y} + p_x \frac{\partial}{\partial p_x} + f \frac{\partial}{\partial p_y} + p_z \frac{\partial}{\partial p_z} + s \frac{\partial}{\partial s} \,.$$

Further work

- Apply this framework to the study of **singular Lagrangians**, both for **action-dependent** and **non-autonomous** systems.
- Use these tools to study the relevant **dynamics** and the **brackets** induced by different geometric structures in more generality.
- Generalization of the framework for pairs of **(n, n+1)-forms**.
- Analyse in detail different situations where the **class is not constant**, and how this may alter the behaviour of the associated geometrical objects.

References

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